Direct Force Compensation-Based Adaptive Hybrid Impedance Control for Dynamic Interaction Environments

Hongjun Xing, *Member, IEEE*, Yuzhe Xu, Yi Xie, Jinbao Chen, Weihua Li, Yanqing Liu, and Liang Ding, *Senior Member, IEEE*

***Abstract*—The primary control objective in robotic interaction tasks has shifted from trajectory tracking accuracy to interaction force regulation. Impedance control enables compliant interaction by shaping the robot's force response through a target impedance model. However, the dynamic response of conventional impedance control is fixed during parameter design, limiting its adaptability to varying interaction environments. To address this issue, a direct force compensation–based adaptive hybrid impedance control (DAHIC) method is proposed. By introducing an adaptive force compensation factor based on the force tracking error, the proposed controller achieves faster response speed while suppressing overshoot and reducing steady-state error, especially in tracking higher-order signals. The allowable range of controller parameters is derived via closed-loop stability analysis, and an adaptive update law is designed accordingly. Furthermore, the proposed adaptive force compensation is integrated with hybrid impedance control to form a six-dimensional task-space compliance controller. The numerical simulations conducted in the single-force subspace, together with the comparative experiments performed in the six-dimensional task space involving constant- and variable-curvature contact surfaces as well as abrupt variations in contact stiffness, collectively validate the improvement in force tracking performance achieved by the proposed algorithm.**

***Note to Practitioners*—In robotic interaction tasks, such as assembly, surface contact, and human–robot collaboration, achieving stable and accurate force regulation in uncertain environments remains challenging for conventional impedance controllers with fixed parameters. This paper proposes a direct force compensation–based adaptive hybrid impedance control method to improve force tracking performance under varying interaction conditions. By adaptively adjusting the force compensation according to tracking error, the controller achieves faster response, reduced overshoot, and improved steady-state accuracy, and can be naturally extended to six-dimensional task-space compliance control. The stability of the closed-loop system is ensured through analytical parameter constraints, providing practical guidance for controller design. Practitioners may find this approach useful for enhancing compliant interaction performance in real-world robotic applications, while practical implementation requires appropriate force sensing, real-time computation, and careful parameter tuning.**

***Index Terms*—Compliance control, impedance control, adaptive control, direct force compensation, robot-environment interaction.**

I. Introduction

IN TRADITIONAL industrial robotic applications, the position-based control paradigm has achieved remarkable success in structured and deterministic environments [1], [2]. However, with the growing demand for human-robot collaboration and operations in unstructured, dynamic scenarios—such as precision assembly and medical rehabilitation—this rigid control approach faces limitations due to its inability to actively adapt to environmental interaction forces. To address this, compliant control has emerged as a key enabling technology. Its core principle lies in regulating the robot's impedance characteristics, either actively or passively, to achieve coordinated force and position adjustment. This approach enables robots to interact safely and adaptively with uncertain environments, significantly enhancing operational robustness, accuracy, and safety. Thus, compliant control provides a fundamental theoretical foundation for the development of next-generation intelligent robots.

Early research focused on mechanical solutions for passive compliance, incorporating elastic elements such as springs, dampers, or airbags at the end-effector (EE) to absorb contact impacts passively. The most representative device is the remote center compliance (RCC) device proposed by Whitney et al. [3], which mechanically realizes a compliance center, allowing the workpiece to undergo slight translation and rotation under contact forces, thereby automatically compensating for assembly misalignment. Wei and Xu [4] further improved and extended this concept to robotic grinding and polishing EEs. Such devices primarily rely on springs or elastic/soft rods to achieve compliance in a passive manner [5].

However, a simple spring alone lacks control capability and cannot adapt to varying compliance requirements under different operating conditions. To address this, Liu et al. [6]

Received 30 December 2025; revised 1 July 2026; accepted 15 August 2026. Date of publication 24 August 2026; date of current version 28 August 2026. This article was recommended for publication by Associate Editor Z. Yao and Editor M. P. Fanti upon evaluation of the reviewers' comments. This work was supported in part by the National Natural Science Foundation of China under Grant 52405025 and Grant 52175007, in part by the Natural Science Foundation of Jiangsu Province under Grant BK20230889, and in part by Shandong Provincial Natural Science Foundation under Grant ZR2024YQ035. *(Corresponding author: Hongjun Xing.)*

Hongjun Xing, Yuzhe Xu, Yi Xie, Jinbao Chen, and Yanqing Liu are with the National Key Laboratory of Aerospace Mechanism, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, China (e-mail: xinghj@nuaa.edu.cn; xuyuzhe@nuaa.edu.cn; yixie10pony@nuaa.edu.cn; chenjbao@nuaa.edu.cn; liuyanqing00@nuaa.edu.cn).

Weihua Li is with the School of Automotive Engineering, Harbin Institute of Technology, Weihai, Weihai 264209, China (e-mail: liweihua@hit.edu.cn).

Liang Ding is with the State Key Laboratory of Robotics and System, Harbin Institute of Technology, Harbin 150001, China (e-mail: liangding@hit.edu.cn).

Digital Object Identifier 10.1109/TASE.2026.3726461

1558-3783 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.

designed interchangeable springs to suit different grinding and polishing objects. Additionally, Hua et al. [7] designed a hydraulically servo-controlled passive compliant joint, which enables the joint to generate passive positional responses upon collision and move in a direction that reduces impact forces. However, due to the fluid-based working medium, pneumatic and hydraulic compliant devices suffer from drawbacks such as hysteresis, slow response, and nonlinearity. Consequently, electric actuation–based alternatives have been proposed to replace pneumatic or hydraulic transmission, improving the responsiveness and output linearity of passive compliant devices. For example, Wang et al. [8] introduced a series elastic actuator (SEA) transmission module driven by a motor, allowing the SEA stiffness to be regulated according to a system stiffness model, thereby achieving fast and approximately linear tunable compliance.

To overcome the inherent limitations of passive compliance, research focus has shifted toward active compliance control. The core concept involves using sensors to perceive interaction information and employing advanced control algorithms to actively regulate the robot's motion or output force. Within this paradigm, two classic theoretical branches have emerged: impedance control [9], [10], [11] and force/position hybrid control [12], [13], [14]. Impedance control does not directly control force or position but instead shapes the robot EE's dynamic behavior as a virtual mass–damper–spring second-order system, indirectly governing the force-position relationship by adjusting its parameters. In contrast, force/position hybrid control adopts a more direct decoupling strategy by decomposing the task space into position-controlled and force-controlled subspaces, enabling precise position tracking or force regulation along different degrees of freedom (DoFs). Hybrid impedance control [15], [16], [17] integrates impedance control with force/position hybrid control, thereby ensuring a controllable and dynamic force-position relationship in all directions while maintaining decoupling between force and position subspaces.

When dealing with operating conditions involving unknown and dynamically changing environmental stiffness, traditional impedance controllers often face challenges. Their performance and reliability are constrained by fixed parameters, which require considerable effort for tuning [18]. In response, research has focused on incorporating adaptive control laws to dynamically optimize model performance. For instance, Seraji and Colbaugh [19] introduced a method that estimates environmental stiffness and position online while selecting appropriate adaptive gains to reduce force errors during interaction with uncertain environments, thereby achieving a certain degree of robustness. Zhang et al. [20] employed an indirect adaptive algorithm to estimate environmental parameters. They utilized a generalized momentum-based disturbance observer to estimate contact forces from torque sensors and proposed an adaptive sliding-mode friction compensation algorithm, which improved the accuracy of position control. Additionally, Duan et al. [21] proposed an adaptive variable-impedance force tracking control method. This approach enables online adjustment of impedance parameters based on tracking errors, facilitating adaptive variable impedance control that compensates for unknown environments and dynamic desired forces.

To enhance the disturbance rejection capability and improve the system robustness of force/position hybrid control strategies, Liang et al. [22] integrated environmental parameter estimation with genetic algorithms. This approach compensates for force errors caused by positional deviations, significantly improving both force tracking accuracy and stability in grinding and polishing operations. Li et al. [23] employed a method combining Generalized Predictive Control (GPC) with neural networks for a six-DoF parallel platform. Through multi-sensor information fusion, they achieved precise force/position tracking on unknown constrained surfaces. Meanwhile, Gracia et al. [24] adopted a task-priority framework, utilizing sliding mode control (SMC) as a high-priority task to enforce constraints on contact force and tool posture. They incorporated an adaptive switching gain to handle abrupt changes in surface stiffness while combining low-priority tasks such as position tracking and configuration maintenance. Their method was validated on a redundant manipulator, demonstrating both effectiveness and low computational complexity. Rojas-Gracia et al. [25] focused on actuator physical constraints and designed an adaptive force/position controller based on a generalized saturation function. By introducing a saturation function to ensure bounded control torque and combining it with a parameter adaptation law for online estimation of gravity and unknown environmental stiffness, they further improved controller robustness under actuator constraints.

Passive compliant mechanisms cannot fully absorb impact energy, while active compliance control places high demands on actuator response speed [26]. To integrate the advantages of both approaches, some studies have combined active and passive compliance, proposing hybrid active-passive compliance control. Xing et al. [27] introduced a compliance control scheme that merges active and passive strategies. This scheme combines a passive compliant EE based on a three-Hooke's-joint parallel structure with active compliance control, adopting a collaborative strategy of "passive compensation for small deviations and active adjustment for large deviations." This effectively addresses EE contact force control and pose tolerance issues in tasks such as door opening and valve rotation. Such a synergistic architecture seeks to achieve a better balance among system stiffness, response speed, control accuracy, and robustness, thereby enabling functional complementarity and performance enhancement at the system level.

With the rapid advancement of artificial intelligence technology, robotic research has entered a new stage of intelligent perception and autonomous decision-making. This progression drives the development of intelligent force control strategies with environmental adaptability and interactive compliance. The focus on compliant force control for industrial robots lies in emulating the adaptive regulatory mechanisms of the human arm in dynamic environments, enabling intelligent responses to external disturbances and robust interactive performance. Consequently, intelligent compliance control that integrates learning and optimization methods holds significant theoretical and practical value [28]. Examples include event-triggered critic learning impedance control that reduces system

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.

overhead [29], inverse reinforcement learning for policy generalization [30], deep reinforcement learning for complex assembly tasks [31], and neural network-based skill learning frameworks for enhanced safety in human-robot collaboration [32].

More recently, reinforcement learning (RL)-based variable impedance control and intelligent hybrid impedance frameworks have attracted increasing attention due to their capability to optimize interaction behaviors in uncertain environments. Buchli et al. [33] proposed a learning-based variable impedance control strategy that adapts impedance behaviors through policy improvement methods. Ding et al. [34] further employed reinforcement learning to optimize impedance parameters for robotic polishing tasks, significantly improving contact stability and surface quality. Sun and Yang [35] introduced an impedance learning-based adaptive interaction controller capable of online estimating environmental impedance profiles for compliant robots. In addition, recent studies have combined deep reinforcement learning with hybrid impedance control to enhance adaptability and robustness in complex interaction scenarios [36].

Although learning-based and optimization-based compliant control strategies have demonstrated strong adaptability and interaction performance, many existing methods rely heavily on large-scale training data, iterative optimization, or deep neural network structures. As a result, their real-time implementation often introduces considerable computational burden and may reduce controller update frequency, especially in high-bandwidth force interaction tasks. Furthermore, guaranteeing closed-loop stability and interpretability remains challenging for many data-driven approaches. In practical industrial applications requiring fast response, stable interaction, and limited computational resources, lightweight adaptive control strategies with explicit stability analysis still possess significant practical advantages.

In addition to the challenges of computational complexity and real-time implementation, compliant control in complex interaction scenarios still faces difficulties in achieving coordinated force-position regulation. In previous studies on compliant control, most methods have focused solely on force tracking along a constant normal direction, with insufficient investigation into the coupling effects between position-based control and force-based control during operations on complex curved surfaces. While existing adaptive hybrid impedance control strategies have enhanced the interactive capabilities of robots in dynamic environments, there remains significant room for improvement in their dynamic performance and force tracking accuracy under coupled motion-contact conditions. Particularly, the interaction between force-controlled and position-controlled subspaces may introduce dynamic coupling effects that degrade force tracking performance, yet this issue has not been sufficiently investigated in existing adaptive hybrid impedance control studies.

In this work, we address the limitation of fixed-parameter impedance controllers in adapting to dynamic environments, aiming to achieve robust and high-precision force tracking performance during physical interaction between robots and their surroundings. The key contributions of this study are summarized as follows:

(1) An adaptive force compensation factor based on force tracking error is proposed. Rather than adjusting impedance parameters, this factor directly modifies the impedance model as an additive term, leading to the design of a direct force compensation–based adaptive impedance control (DAIC).

(2) Through closed-loop stability analysis, the allowable parameter range for DAIC is derived. An adaptive update law is designed for the parameters, following the control principle of "low gain under large errors and high gain under small errors."

(3) The DAIC is integrated into a hybrid impedance control framework, forming the direct force compensation–based hybrid adaptive impedance control (DAHIC) scheme. Experimental analysis is conducted to investigate how introducing position-based impedance control in other directions affects force tracking performance.

The remainder of this paper is organized as follows: Section II introduces hybrid impedance control. Section III elaborates on the design of DAIC, including stability and steady-state error analyses. Section IV presents numerical simulations and comparative experiments to evaluate the force tracking performance of the proposed algorithm and provides parameter guidelines. Finally, Section V presents the conclusions.

II. Hybrid Impedance Control Framework

This section first reviews the design methodology of the standard force-based impedance controller. It then introduces the design of the position-based impedance controller in other directions, presenting the hybrid impedance control framework. This serves as the theoretical basis for the subsequent introduction of the adaptive hybrid impedance controller.

A. Force-Based Impedance Control

Impedance control aims to shape the EE interaction behavior such that, in a chosen task-space direction, the local input-output relation can be approximated by an equivalent second-order mass-spring-damper system. The linear second-order impedance model in a single direction can be described by the transfer function $V(s) = 1/\left(M_d s^2 + B_d s + K_d\right)$; thus, the dynamic relationship between the force tracking error ΔF and the position deviation E can be expressed as:

$$\Delta F = F_e - F_d = M_d \ddot{E} + B_d \dot{E} + K_d E \tag{1}$$

where F_e and F_d denote the measured and desired contact forces, respectively. $E = x - x_d$ represents the position error between the actual EE position x and the desired position x_d. M_d, B_d, and K_d are the desired inertia, damping, and stiffness coefficients, respectively, which define a virtual second-order dynamic system describing the coupling between force and motion deviations. The impedance control model converts the force tracking error ΔF into a position deviation E, and E is used to modify the command tracking signal $x_h = x_d + E$. Here, x_h (or $\boldsymbol{x}_h$ in the multidimensional case) denotes the position command generated by the outer loop and sent to the inner-loop position servo controller.

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.

For the purpose of explaining the basic force-based impedance mechanism in a single Cartesian direction, the local interaction between the EE and the environment is approximated by a linear elastic contact model. Accordingly, the contact force in that direction can be expressed as:

$$F_e = K_e(x_o - x) \tag{2}$$

where x_o denotes the environment surface position corresponding to the zero-force contact point, and K_e is the equivalent local contact stiffness in the considered direction. Here, the above scalar relation is introduced only to illustrate the force-regulation principle of the standard impedance controller. In the subsequent hybrid impedance formulation, the interaction model will be extended to a multidimensional task-space description to account for cross-axis coupling and environment anisotropy.

For low-velocity interaction tasks and sufficiently fast inner-loop position servo behavior, the commanded position can be approximated by the actual position in the nominal analysis, *i.e.*, $x_h \approx x$. Under this simplifying assumption, substituting the above contact model into the impedance equation yields:

$$\Delta F = K_e(x_o - x_d - E) - F_d \tag{3}$$

This approximation is introduced here only to reveal the basic force-regulation mechanism of the standard impedance controller. In the subsequent hybrid impedance analysis, the inner-loop position servo is modeled explicitly as a finite-bandwidth first-order system.

Assuming a constant desired force, and taking the Laplace transform of (3). By substituting $E(s) = V(s)\Delta F$ and $V(s) = 1/\left(M_d s^2 + B_d s + K_d\right)$, the force tracking error in the Laplace domain is obtained as:

$$\Delta F(s) = \frac{[K_e(x_o - x_d) - F_d](M_d s^2 + B_d s + K_d)}{M_d s^3 + B_d s^2 + (K_d + K_e)s} \tag{4}$$

Thus, the steady-state force tracking error is given by:

$$\Delta F_{ss} = \lim_{s\to 0} s\Delta F(s) = \frac{K_d}{K_d + K_e}\left[K_e(x_o - x_d) - F_d\right] \tag{5}$$

Under the above scalar contact model and nominal servo approximation, the steady-state force tracking error converges to zero if at least one of the following conditions is satisfied:

$$\begin{cases} K_d = 0 \\ x_d = x_o - \dfrac{F_d}{K_e} \end{cases} \tag{6}$$

To satisfy the second condition in (6), it is necessary to obtain not only the contact surface position x_o but also an accurate estimate of the equivalent local contact stiffness K_e in the considered direction. However, in practical interaction scenarios, the environmental stiffness may vary significantly across different environments or even within the same environment, making precise estimation difficult. To avoid explicit reliance on uncertain environmental parameters in the scalar nominal analysis, the stiffness term is set to $K_d = 0$, which allows the steady-state force tracking error to vanish under the above assumptions. Accordingly, the impedance model in (1) is simplified as:

$$\Delta F = M_d\ddot{E} + B_d\dot{E} \tag{7}$$

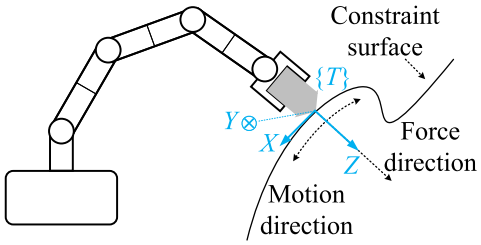


Fig. 1. Schematic diagram of the task coordinate frame.

B. Hybrid Impedance Control

In interactive tasks such as dust removal and grinding, the integration of force control and position control is often essential. Hybrid impedance control combines impedance control with force/position hybrid control, allowing for the independent assignment of force or position control modes to different DoFs in the task space. This facilitates coordinated and compliant regulation of both the EE motion and contact forces.

To implement hybrid impedance control, it is first necessary to establish a task coordinate frame $\{T\}$ that is aligned with the task geometry. The Z-axis of this coordinate frame is aligned with the normal direction of the contact surface, while the X- and Y-axes lie in the tangential plane, as illustrated in Fig. 1.

Subsequently, a selection matrix $\boldsymbol{S}$ is defined to determine which directions in $\{T\}$ are assigned to force control and which to position control. The selection matrix $\boldsymbol{S}$ is a diagonal matrix, where $s_i = 1$ indicates that the i-th DoF is governed by position control, and $s_i = 0$ indicates force control.

In the research presented in this chapter, a six-dimensional Cartesian task space is employed. Force-based impedance control is implemented along the normal direction of the workpiece surface, while position-based impedance control is applied to the tangential directions. Additionally, position-based impedance control is used for orientation regulation to ensure that the EE remains perpendicular to the workpiece surface. Accordingly, the selection matrix is configured as:

$$\boldsymbol{S} = \mathrm{diag}(1, 1, 0, 1, 1, 1) \tag{8}$$

where $\mathrm{diag}(\cdot)$ represents a diagonal matrix whose entries correspond to the values along its main diagonal.

Position-based impedance control ensures compliant operation of the system by tracking desired position commands while incorporating force feedback. The differences in control command convergence between position-based and force-based impedance control lead to distinct structural configurations [15]. A typical formulation for a position-based impedance controller is given by (9), which corresponds to the special case of (1) with zero desired force.

$$\boldsymbol{F}_e = \boldsymbol{M}_d(\ddot{\boldsymbol{x}} - \ddot{\boldsymbol{x}}_d) + \boldsymbol{B}_d(\dot{\boldsymbol{x}} - \dot{\boldsymbol{x}}_d) + \boldsymbol{K}_d(\boldsymbol{x} - \boldsymbol{x}_d) \tag{9}$$

For impedance control in the attitude DoFs, analogous to impedance control in the positional DoFs, torque $\boldsymbol{\tau}$ and angular velocity $\boldsymbol{\omega}$ are used to describe rotational motion. However, the integral of $\boldsymbol{\omega}$ does not provide a physically consistent

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.

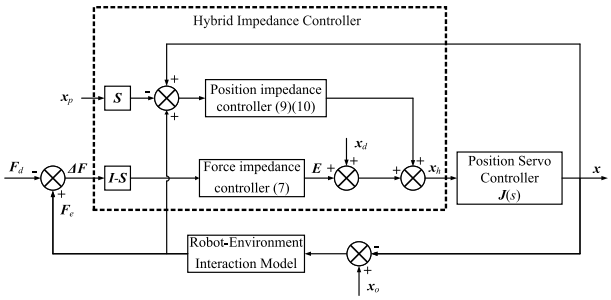


Fig. 2. Block diagram of the hybrid impedance control system.

representation and is therefore unsuitable as a quantity for describing rotational displacement. Similarly, the commonly used Euler angle representation suffers from kinematic singularities and is not geometrically consistent with $\boldsymbol{\omega}$, which limits its applicability in control design.

To address these limitations, unit quaternions $\boldsymbol{Q} = \{\eta, \boldsymbol{\varepsilon}\} \in \mathbb{R}^4$ are adopted to represent rotational displacement, where η denotes the scalar part and $\boldsymbol{\varepsilon} = [\epsilon_x, \epsilon_y, \epsilon_z]^{\mathrm{T}} \in \mathbb{R}^3$ denotes the vector part associated with the rotation axis. Due to their singularity-free representation and direct kinematic relationship with angular velocity $\boldsymbol{\omega}$, unit quaternions are widely used for describing rotational motion [37]. Accordingly, the impedance model for attitude control is formulated as follows:

$$\boldsymbol{\tau} = \boldsymbol{M}_o\dot{\boldsymbol{\omega}} + \boldsymbol{B}_o\boldsymbol{\omega} + \boldsymbol{K}_o(\boldsymbol{\varepsilon}_c - \boldsymbol{\varepsilon}_d) \tag{10}$$

where $\boldsymbol{M}_o$, $\boldsymbol{B}_o$, and $\boldsymbol{K}_o$ are the desired inertia, damping, and stiffness matrices of the attitude impedance, respectively. $\boldsymbol{Q}_c = \{\eta_c, \boldsymbol{\varepsilon}_c\}$ and $\boldsymbol{Q}_d = \{\eta_d, \boldsymbol{\varepsilon}_d\}$ denote the current and desired attitude unit quaternion representations, with η_c and η_d being the scalar components, and $\boldsymbol{\varepsilon}_c$ and $\boldsymbol{\varepsilon}_d$ the vector components. The expression for $\boldsymbol{K}_o$ is given by:

$$\boldsymbol{K}_o = 2\boldsymbol{V}^{\mathrm{T}}(\eta_c, \boldsymbol{\varepsilon}_c)\boldsymbol{K}_o' \tag{11}$$

$$\boldsymbol{V} = \eta_c\boldsymbol{I} - \boldsymbol{\Omega}(\boldsymbol{\varepsilon}_c) \tag{12}$$

where $\boldsymbol{K}_o'$ is the intermediate stiffness matrix for attitude impedance, and $\boldsymbol{\Omega}(\boldsymbol{\varepsilon}) = [\boldsymbol{\varepsilon}]\times$ is the skew-symmetric matrix representing the vector cross-product operation.

In conclusion, the block diagram of the hybrid impedance control system is illustrated in Fig. 2. In this framework, $\boldsymbol{x}_p$ denotes the position input, and $\boldsymbol{x}_h$ represents the hybrid impedance control command synthesized from the position-based impedance controller and the force-based impedance controller. This control command is then used as the reference input to the position servo controller. The updated robot position, together with the six-dimensional interaction force $\boldsymbol{F}_e$ measured at the EE, forms a closed-loop feedback system.

It should be emphasized that the hybrid impedance framework in Fig. 2 is formulated in the task space of the EE and is intended for multi-DoF physical interaction. Therefore, although the force regulation objective is assigned only to the force-controlled subspace, the actual closed-loop behavior is influenced by the coupled robot dynamics, finite-bandwidth position servo behavior, and multidimensional contact interactions. For this reason, the adaptive controller developed in the next section is analyzed directly in a multiple-input-multiple-output (MIMO) task-space setting rather than only in a scalar force-control direction.

III. Direct Force Compensation–Based Adaptive Hybrid Impedance Control

In practical robot-environment interaction, the design of an adaptive force controller is complicated by several factors. First, the environment stiffness is generally unknown and may vary significantly across different contact conditions, which limits the effectiveness of model-dependent force regulation strategies. Second, although increasing the force-feedback effect can improve response speed, it often leads to larger overshoot and oscillation, especially in the presence of servo lag and force-measurement noise. Third, for multi-DoF manipulators, the interaction process is inherently coupled, so a purely scalar or idealized kinematic treatment is insufficient to characterize the actual closed-loop behavior. To address these issues, a DAIC strategy is developed in this section. The proposed method preserves the nominal impedance structure while introducing a bounded adaptive direct force-compensation term, and its closed-loop behavior is analyzed in a task-space MIMO framework.

A. Task-Space Dynamic Formulation

To analyze the proposed controller in a physically meaningful manner, the closed-loop robot–environment interaction is formulated in the task space of the EE. Let $\boldsymbol{q} \in \mathbb{R}^n$ denote the joint coordinates of the manipulator and $\boldsymbol{x} \in \mathbb{R}^6$ denote the Cartesian pose of the EE in the task frame $\{T\}$. The rigid-body dynamics of the manipulator are described by:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau}_c + \boldsymbol{\tau}_e + \boldsymbol{\tau}_d \tag{13}$$

where $\boldsymbol{M}(\boldsymbol{q})$ is the joint-space inertia matrix, $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ is the Coriolis/centrifugal matrix, $\boldsymbol{g}(\boldsymbol{q})$ is the gravity vector, $\boldsymbol{q}$ is the joint position vector, $\boldsymbol{\tau}_c$ is the commanded joint torque, $\boldsymbol{\tau}_e$ is the generalized torque induced by the contact wrench, and $\boldsymbol{\tau}_d$ denotes lumped disturbances including friction, parameter uncertainty, and unmodeled dynamics.

Using standard operational-space mapping, the corresponding task-space dynamics can be written in the compact form:

$$\boldsymbol{M}_x(\boldsymbol{q})\ddot{\boldsymbol{x}} + \boldsymbol{C}_x(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{x}} + \boldsymbol{g}_x(\boldsymbol{q}) = \boldsymbol{u} + \boldsymbol{F}_e + \boldsymbol{d}_x \tag{14}$$

where $\boldsymbol{M}_x(\boldsymbol{q}) \in \mathbb{R}^{6\times6}$ is the task-space inertia matrix, $\boldsymbol{C}_x(\boldsymbol{q}, \dot{\boldsymbol{q}})$ is the corresponding Coriolis/centrifugal term, $\boldsymbol{g}_x(\boldsymbol{q})$ is the task-space gravity term, $\boldsymbol{u}$ is the control input generated by the hybrid impedance controller, $\boldsymbol{F}_e \in \mathbb{R}^6$ is the measured interaction wrench at the EE, and $\boldsymbol{d}_x$ is the lumped task-space disturbance.

To describe hybrid force/position interaction, the selection matrix $\boldsymbol{S} \in \mathbb{R}^{6\times6}$ introduced in Section II is used to define the position-controlled and force-controlled subspaces:

$$\boldsymbol{P}_p = \boldsymbol{S}, \qquad \boldsymbol{P}_f = \boldsymbol{I} - \boldsymbol{S} \tag{15}$$

where $\boldsymbol{P}_p$ and $\boldsymbol{P}_f$ are complementary projection matrices. Accordingly, the task-space motion and interaction wrench can be decomposed as:

$$\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_f, \qquad \boldsymbol{F}_e = \boldsymbol{F}_{e,p} + \boldsymbol{F}_{e,f} \tag{16}$$

with $\boldsymbol{x}_p = \boldsymbol{P}_p\boldsymbol{x}$, $\boldsymbol{x}_f = \boldsymbol{P}_f\boldsymbol{x}$, $\boldsymbol{F}_{e,p} = \boldsymbol{P}_p\boldsymbol{F}_e$, and $\boldsymbol{F}_{e,f} = \boldsymbol{P}_f\boldsymbol{F}_e$.

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.

In practical contact tasks, the environment is generally nonlinear and anisotropic. Nevertheless, around the operating point and for low-speed interaction, the contact behavior can be locally approximated by:

$$\boldsymbol{F}_e = \boldsymbol{K}_e(\boldsymbol{x}_o - \boldsymbol{x}) + \boldsymbol{D}_e(\dot{\boldsymbol{x}}_o - \dot{\boldsymbol{x}}) + \boldsymbol{\Delta}_e(\boldsymbol{x}, \dot{\boldsymbol{x}}, t) \tag{17}$$

where $\boldsymbol{K}_e \in \mathbb{R}^{6\times6}$ and $\boldsymbol{D}_e \in \mathbb{R}^{6\times6}$ denote the local environment stiffness and damping matrices, respectively, $\boldsymbol{x}_o$ is the local zero-force contact equilibrium, and $\boldsymbol{\Delta}_e$ represents the residual nonlinear contact uncertainty. Therefore, the multidimensional coupling induced by robot dynamics and environment anisotropy is explicitly retained in the MIMO formulation.

B. Direct Force Compensation Adaptive Hybrid Impedance Controller

The hybrid impedance controller is constructed by assigning position-based impedance control to the position-controlled subspace and DAIC to the force-controlled subspace. The total task-space command is given by:

$$\boldsymbol{u} = \boldsymbol{P}_p\boldsymbol{u}_p + \boldsymbol{P}_f\boldsymbol{u}_f. \tag{18}$$

For the position-controlled subspace, the desired impedance behavior is defined as:

$$\boldsymbol{u}_p = \boldsymbol{M}_{dp}(\ddot{\boldsymbol{x}}_d - \ddot{\boldsymbol{x}}) + \boldsymbol{B}_{dp}(\dot{\boldsymbol{x}}_d - \dot{\boldsymbol{x}}) + \boldsymbol{K}_{dp}(\boldsymbol{x}_d - \boldsymbol{x}) \tag{19}$$

where $\boldsymbol{M}_{dp}$, $\boldsymbol{B}_{dp}$, and $\boldsymbol{K}_{dp}$ are symmetric positive-definite desired inertia, damping, and stiffness matrices for the position-controlled subspace.

For the force-controlled subspace, let

$$\Delta\boldsymbol{F} = \boldsymbol{F}_{e,f} - \boldsymbol{F}_d \tag{20}$$

denote the force tracking error, where $\boldsymbol{F}_d$ is the desired contact wrench in the force-controlled subspace. The proposed DAIC law is formulated as:

$$\Delta\boldsymbol{F} = \boldsymbol{M}_{df}\ddot{\boldsymbol{E}}_f + \boldsymbol{B}_{df}\dot{\boldsymbol{E}}_f + \boldsymbol{\Lambda}(t) \tag{21}$$

where $\boldsymbol{E}_f$ is the position correction generated by the force controller, $\boldsymbol{M}_{df}$ and $\boldsymbol{B}_{df}$ are symmetric positive-definite desired inertia and damping matrices in the force-controlled subspace, and $\boldsymbol{\Lambda}(t)$ is the adaptive direct force compensation term.

The adaptive compensation term is defined as:

$$\boldsymbol{\Lambda}(t) = \alpha\boldsymbol{\Lambda}(t-T) + \boldsymbol{B}_\beta(t)\left[\boldsymbol{F}_d(t-T) - \boldsymbol{F}_{e,f}(t-T)\right] \tag{22}$$

where T is the control period, $\alpha \in (0, 1]$ is the attenuation factor of the accumulated compensation, and $\boldsymbol{B}_\beta(t)$ is a diagonal adaptive gain matrix:

$$\boldsymbol{B}_\beta(t) = \mathrm{diag}\left(\beta_1(t), \beta_2(t), \ldots, \beta_m(t)\right) \tag{23}$$

with $m = \mathrm{rank}(\boldsymbol{P}_f)$ denoting the dimension of the force-controlled subspace.

Unlike conventional adaptive impedance strategies that update impedance parameters online, the proposed DAIC preserves the nominal impedance matrices $\boldsymbol{M}_{df}$ and $\boldsymbol{B}_{df}$ and instead introduces a direct compensation channel in the force subspace. This structure improves force-transient regulation while retaining the physical interpretability of the nominal impedance model.

Following the design principle used in the scalar case, each adaptive gain $\beta_i(t)$ is selected according to:

$$\beta_i(t) = \frac{1}{e^{k_{1i}|\Delta F_i|} + e^{k_{2i}|\Delta\dot{F}_i|} + \bar{\beta}_i}, \qquad i = 1, \ldots, m \tag{24}$$

where $k_{1i} > 0$ and $k_{2i} > 0$ are design gains, and $\bar{\beta}_i > 0$ is selected such that $0 < \beta_i(t) \le \beta_{i,\max}$ for all time. In this way, the adaptive gain remains bounded and can be incorporated into the multidimensional closed-loop stability analysis. In implementation, $\Delta\dot{F}_i$ is computed using a filtered differentiation scheme to avoid amplification of sensor noise.

C. Finite-Bandwidth Position Servo Model

In many impedance-control studies, the inner-loop position servo dynamics are approximated as ideal, namely $J(s) = 1$. This approximation is convenient for analysis but may be inaccurate in practical industrial manipulators due to finite servo bandwidth, actuator lag, joint flexibility, and unmodeled high-frequency dynamics. To account for these effects, the inner-loop position servo is modeled here as a first-order stable system:

$$\dot{\boldsymbol{x}} = -\boldsymbol{\Omega}_p\left(\boldsymbol{x} - \boldsymbol{x}_h\right) + \boldsymbol{d}_s \tag{25}$$

where $\boldsymbol{x}_h$ denotes the hybrid impedance command generated by the outer-loop controller, $\boldsymbol{\Omega}_p = \mathrm{diag}(\omega_{p,1}, \ldots, \omega_{p,6})$ is the diagonal bandwidth matrix of the position servo, and $\boldsymbol{d}_s$ is the lumped servo-side uncertainty including unmodeled inner-loop dynamics and tracking bias.

Equivalently, in the Laplace domain, the servo dynamics can be expressed as:

$$\boldsymbol{J}(s) = \mathrm{diag}\left(\frac{\omega_{p,1}}{s+\omega_{p,1}}, \ldots, \frac{\omega_{p,6}}{s+\omega_{p,6}}\right). \tag{26}$$

Therefore, the ideal assumption $J(s) = 1$ used in simplified analyses can be interpreted as the limiting case of (26) when $\omega_{p,i} \to \infty$. For finite $\omega_{p,i}$, the additional phase lag reduces the effective stability margin and introduces a coupling between the outer-loop adaptive impedance dynamics and the inner-loop servo tracking error.

Define the servo tracking error as:

$$\tilde{\boldsymbol{x}} = \boldsymbol{x} - \boldsymbol{x}_h. \tag{27}$$

Then, from (25), its dynamics satisfy:

$$\dot{\tilde{\boldsymbol{x}}} = -\boldsymbol{\Omega}_p\tilde{\boldsymbol{x}} + \boldsymbol{d}_s - \dot{\boldsymbol{x}}_h. \tag{28}$$

Equation (28) shows that the finite-bandwidth servo behaves as an exponentially stable subsystem driven by the outer-loop command rate and the servo disturbance. Consequently, if the outer-loop command $\boldsymbol{x}_h$ varies sufficiently slowly relative to the inner-loop bandwidth, the servo tracking error remains bounded and small. This property provides the basis for the subsequent cascade stability analysis.

Based on the above design considerations, the flowchart of the DAIC algorithm is illustrated in Fig. 3.

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.

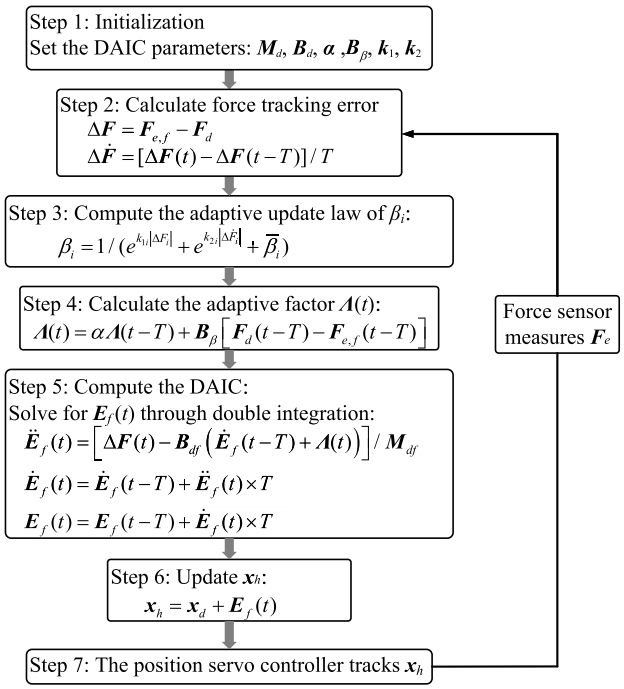


Fig. 3. Flowchart of the DAIC algorithm.

D. MIMO Closed-Loop Stability Analysis

In this subsection, the closed-loop stability of the proposed DAHIC is analyzed in the MIMO task-space setting. The analysis explicitly considers robot dynamics, multidimensional coupling, local environment uncertainty, and finite-bandwidth position servo effects.

Define the task-space tracking error as:

$$\boldsymbol{e}_x = \boldsymbol{x} - \boldsymbol{x}_d, \qquad \dot{\boldsymbol{e}}_x = \dot{\boldsymbol{x}} - \dot{\boldsymbol{x}}_d \tag{29}$$

and the force tracking error in the force-controlled subspace is:

$$\boldsymbol{e}_f = \boldsymbol{F}_{e,f} - \boldsymbol{F}_d. \tag{30}$$

Since the adaptive compensation term in (22) is updated in discrete time, a continuous-time approximation is introduced to facilitate analysis. For sufficiently small sampling period T, the incremental relation

$$\boldsymbol{\Lambda}(t) - \boldsymbol{\Lambda}(t-T) \approx T\dot{\boldsymbol{\Lambda}}(t) \tag{31}$$

yields the approximate compensation dynamics

$$\dot{\boldsymbol{\Lambda}} = -\frac{1-\alpha}{T}\boldsymbol{\Lambda} + \frac{1}{T}\boldsymbol{B}_\beta(t)\boldsymbol{e}_f + \boldsymbol{r}_T(t) \tag{32}$$

where $\boldsymbol{r}_T(t)$ denotes the residual approximation error due to sampling and neglected higher-order terms. Under sufficiently fast sampling, $\boldsymbol{r}_T(t)$ is bounded.

Substituting the hybrid controller into the task-space dynamics (14), the closed-loop interaction dynamics can be expressed in the compact form as:

$$\begin{aligned}&\boldsymbol{M}_x(\boldsymbol{q})\ddot{\boldsymbol{e}}_x + \boldsymbol{C}_x(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{e}}_x + \boldsymbol{K}_{cl}\boldsymbol{e}_x + \boldsymbol{D}_{cl}\dot{\boldsymbol{e}}_x = \\ &\boldsymbol{W}_\Lambda\boldsymbol{\Lambda} + \boldsymbol{\Delta}(\boldsymbol{\chi},t)\end{aligned} \tag{33}$$

where $\boldsymbol{K}_{cl}$ and $\boldsymbol{D}_{cl}$ are the effective closed-loop stiffness and damping matrices induced by the position- and force-subspace impedance laws, $\boldsymbol{W}_\Lambda$ is the input distribution matrix of the adaptive compensation term, and $\boldsymbol{\Delta}(\boldsymbol{\chi},t)$ is the lumped perturbation term depending on the augmented state $\boldsymbol{\chi}$. The latter includes residual dynamic coupling, environment-model mismatch, servo-tracking error, sensor noise, and external disturbances.

The following assumptions are introduced.

Assumption 1: The matrices $\boldsymbol{M}_x(\boldsymbol{q})$, $\boldsymbol{C}_x(\boldsymbol{q},\dot{\boldsymbol{q}})$ satisfy the standard robot dynamic properties, namely $\boldsymbol{M}_x(\boldsymbol{q})$ is symmetric positive definite and $\dot{\boldsymbol{M}}_x - 2\boldsymbol{C}_x$ is skew-symmetric.

Assumption 2: The environment uncertainty, servo disturbance, and lumped perturbation are bounded, *i.e.*,

$$\|\boldsymbol{\Delta}(\boldsymbol{\chi},t)\| \le \rho_1\|\boldsymbol{e}_x\| + \rho_2\|\dot{\boldsymbol{e}}_x\| + \rho_3\|\boldsymbol{\Lambda}\| + \rho_4\|\tilde{\boldsymbol{x}}\| + \bar{d} \tag{34}$$

for some nonnegative constants $\rho_1,\rho_2,\rho_3,\rho_4,\bar{d}$.

Assumption 3: The adaptive gains remain bounded, *i.e.*, there exist constants $\beta_{i,\max} > 0$ such that $0 < \beta_i(t) \le \beta_{i,\max}$ for all i and all time.

Consider the Lyapunov candidate:

$$\begin{aligned}V = &\frac{1}{2}\dot{\boldsymbol{e}}_x^{\mathrm{T}}\boldsymbol{M}_x(\boldsymbol{q})\dot{\boldsymbol{e}}_x + \frac{1}{2}\boldsymbol{e}_x^{\mathrm{T}}\boldsymbol{K}_{cl}\boldsymbol{e}_x \\ &+ \frac{1}{2}\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{P}_\Lambda\boldsymbol{\Lambda} + \frac{1}{2}\tilde{\boldsymbol{x}}^{\mathrm{T}}\boldsymbol{P}_s\tilde{\boldsymbol{x}}\end{aligned} \tag{35}$$

where $\boldsymbol{P}_\Lambda$ and $\boldsymbol{P}_s$ are symmetric positive-definite weighting matrices.

Taking the time derivative of (35) gives:

$$\begin{aligned}\dot{V} = &\dot{\boldsymbol{e}}_x^{\mathrm{T}}\boldsymbol{M}_x\ddot{\boldsymbol{e}}_x + \frac{1}{2}\dot{\boldsymbol{e}}_x^{\mathrm{T}}\dot{\boldsymbol{M}}_x\dot{\boldsymbol{e}}_x + \boldsymbol{e}_x^{\mathrm{T}}\boldsymbol{K}_{cl}\dot{\boldsymbol{e}}_x \\ &+ \boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{P}_\Lambda\dot{\boldsymbol{\Lambda}} + \tilde{\boldsymbol{x}}^{\mathrm{T}}\boldsymbol{P}_s\dot{\tilde{\boldsymbol{x}}}\end{aligned} \tag{36}$$

Substituting (33) into (36), and using the skew-symmetry property of $\dot{\boldsymbol{M}}_x - 2\boldsymbol{C}_x$ together with the symmetry of $\boldsymbol{K}_{cl}$, yields:

$$\dot{V} = -\dot{\boldsymbol{e}}_x^{\mathrm{T}}\boldsymbol{D}_{cl}\dot{\boldsymbol{e}}_x + \dot{\boldsymbol{e}}_x^{\mathrm{T}}\boldsymbol{W}_\Lambda\boldsymbol{\Lambda} + \dot{\boldsymbol{e}}_x^{\mathrm{T}}\boldsymbol{\Delta} + \boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{P}_\Lambda\dot{\boldsymbol{\Lambda}} + \tilde{\boldsymbol{x}}^{\mathrm{T}}\boldsymbol{P}_s\dot{\tilde{\boldsymbol{x}}}. \tag{37}$$

Then, substituting (32) and (28) into (37) gives:

$$\begin{aligned}\dot{V} = &-\dot{\boldsymbol{e}}_x^{\mathrm{T}}\boldsymbol{D}_{cl}\dot{\boldsymbol{e}}_x - \frac{1-\alpha}{T}\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{P}_\Lambda\boldsymbol{\Lambda} - \tilde{\boldsymbol{x}}^{\mathrm{T}}\boldsymbol{P}_s\boldsymbol{\Omega}_p\tilde{\boldsymbol{x}} \\ &+ \dot{\boldsymbol{e}}_x^{\mathrm{T}}\boldsymbol{W}_\Lambda\boldsymbol{\Lambda} + \dot{\boldsymbol{e}}_x^{\mathrm{T}}\boldsymbol{\Delta} + \frac{1}{T}\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{P}_\Lambda\boldsymbol{B}_\beta\boldsymbol{e}_f \\ &+ \boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{P}_\Lambda\boldsymbol{r}_T + \tilde{\boldsymbol{x}}^{\mathrm{T}}\boldsymbol{P}_s\boldsymbol{d}_s - \tilde{\boldsymbol{x}}^{\mathrm{T}}\boldsymbol{P}_s\dot{\boldsymbol{x}}_h.\end{aligned} \tag{38}$$

By **Assumptions 2–3**, the boundedness of $\boldsymbol{B}_\beta$, and repeated application of Young's inequality, all cross terms in (38) can be upper-bounded by quadratic terms in $\|\boldsymbol{e}_x\|$, $\|\dot{\boldsymbol{e}}_x\|$, $\|\boldsymbol{\Lambda}\|$, and $\|\tilde{\boldsymbol{x}}\|$, together with bounded residual terms depending on $\|\boldsymbol{r}_T\|$, $\|\boldsymbol{d}_s\|$, $\|\dot{\boldsymbol{x}}_h\|$, and the contact uncertainty. Moreover, since the force tracking error is generated by the task-space motion error, servo error, and bounded contact uncertainty, there exist positive constants $c_{f1},c_{f2},c_{f3},c_{f4}$ such that:

$$\|\boldsymbol{e}_f\|^2 \le c_{f1}\|\boldsymbol{e}_x\|^2 + c_{f2}\|\dot{\boldsymbol{e}}_x\|^2 + c_{f3}\|\tilde{\boldsymbol{x}}\|^2 + c_{f4}. \tag{39}$$

Therefore, by choosing the desired damping sufficiently large, the servo bandwidth sufficiently high, and the sampling period sufficiently small, one obtains:

$$\begin{aligned}\dot{V} \le &-c_1\|\dot{\boldsymbol{e}}_x\|^2 - c_2\|\boldsymbol{e}_x\|^2 - c_3\|\boldsymbol{\Lambda}\|^2 - c_4\|\tilde{\boldsymbol{x}}\|^2 \\ &+ \nu_1\|\boldsymbol{r}_T\|^2 + \nu_2\bar{d}^2,\end{aligned} \tag{40}$$

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.

for some positive constants $c_1, c_2, c_3, c_4, \nu_1, \nu_2$, where $\bar{d}$ collects the bounded disturbance contributions including servo uncertainty and contact model mismatch. Hence, the closed-loop system is uniformly ultimately bounded.

Therefore, the following result holds.

Proposition 1: Consider the MIMO robot–environment interaction system described by (14)–(28). Suppose **Assumptions 1–3** hold, and suppose the desired impedance matrices $\boldsymbol{M}_{dp}$, $\boldsymbol{B}_{dp}$, $\boldsymbol{K}_{dp}$, $\boldsymbol{M}_{df}$, and $\boldsymbol{B}_{df}$ are symmetric positive definite. Then, for sufficiently small sampling period T and sufficiently large inner-loop bandwidth $\boldsymbol{\Omega}_p$, the closed-loop error signals $\boldsymbol{e}_x$, $\dot{\boldsymbol{e}}_x$, $\boldsymbol{\Lambda}$, and $\tilde{\boldsymbol{x}}$ are uniformly ultimately bounded. Moreover, in the disturbance-free ideal case $\boldsymbol{\Delta} \equiv 0$, $\boldsymbol{r}_T \equiv 0$, and $\boldsymbol{d}_s \equiv 0$, the error system asymptotically converges to the origin.

Proposition 1 indicates that the proposed controller remains stable in the presence of multidimensional coupling and finite-bandwidth inner-loop dynamics, provided that the residual uncertainties are bounded and the servo bandwidth is sufficiently higher than the outer-loop impedance dynamics. In this sense, the ideal assumption $J(s) = 1$ corresponds only to a limiting nominal case, whereas the more realistic model (26) leads to a practically meaningful bounded-stability result for the full MIMO closed-loop system.

E. Robustness Discussion

The above analysis shows that the proposed controller should be interpreted as an outer-loop adaptive impedance strategy with bounded-disturbance robustness, rather than as an exact full-order inverse-dynamics controller. Its robustness properties can be summarized as follows.

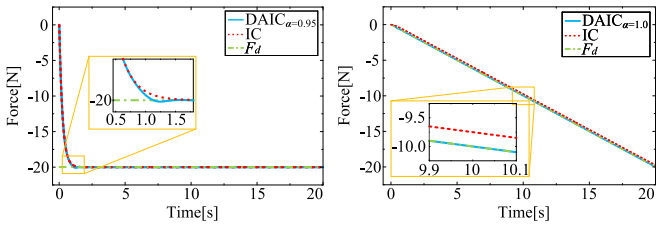
First, the proposed DAIC law does not require explicit online estimation of the environment stiffness matrix $\boldsymbol{K}_e$. Therefore, abrupt changes in contact stiffness do not directly invalidate the controller structure, although they do affect the transient tracking error through the perturbation term $\boldsymbol{\Delta}(\chi, t)$.

Second, the finite-bandwidth position servo model (25) shows that a reduction in servo bandwidth increases the servo tracking error $\tilde{\boldsymbol{x}}$, thereby reducing the effective damping margin of the outer-loop impedance dynamics. Consequently, the admissible range of adaptive gains becomes more conservative when the inner-loop bandwidth is low.

Third, the derivative term used in the adaptive gain law (24) is sensitive to force sensor noise. For this reason, the practical implementation employs filtered differentiation or smoothed discrete difference estimation. Under bounded measurement noise, the resulting gain variation remains bounded and does not destroy the boundedness result in Proposition 1.

Finally, the local environment model (17) should be understood as a low-velocity local linearization. For strongly nonlinear contact conditions, actuator saturation, or severe structural flexibility, the perturbation term may increase substantially, leading to a larger ultimate tracking-error bound. This limitation is inherent to the present low-frequency task-space stability analysis and should be taken into account in practical deployment.

To complement the above MIMO bounded-stability analysis, Appendix provides a scalar steady-state tracking interpretation of the proposed DAIC law under simplified nominal assumptions. This supplementary analysis is intended only to offer additional intuition regarding the low-frequency force-tracking characteristics of the compensation mechanism.



(a) Step response (b) Ramp response

Fig. 4. Comparison of force tracking performance between DAIC and IC under different input signals.

IV. Simulations and Experiments

This section evaluates the performance of the proposed algorithm and provides guidelines for controller parameter design through numerical simulations and experimental validation. Numerical simulations are conducted to verify the convergence properties of the algorithm and analyze the influence of controller parameters on force tracking performance. Subsequently, comparative experiments are performed in a six-dimensional task space to demonstrate the effectiveness of the proposed DAHIC algorithm and to further investigate the impact of position-based impedance control parameters on force tracking accuracy. In both simulations and experiments, a position servo controller based on Hierarchical Quadratic Programming (HQP) [38], [39] is employed to enforce joint-level physical constraints. The control frequency is set to 100 Hz.

A. Numerical Simulation Analysis of DAIC

To validate the effectiveness of the DAIC method while eliminating uncertainties caused by noise in real physical environments, this subsection focuses on force tracking performance in a single force-subspace direction. A numerical comparative simulation is conducted to evaluate DAIC against the standard impedance controller in (7), comparing their force tracking performance under ideal conditions.

The force tracking performance of the two controllers is first evaluated in a static environment, where the environmental stiffness is set to $K_e = 400$ N/m, and the impedance parameters are $M_d = 1$ Ns^2/m, $B_d = 100$ Ns/m. The parameters of the adaptive update law for β are set as $k_1 = k_2 = 1$. The simulation results are presented in Fig. 4, where "IC" denotes the standard impedance controller, and the parameters of the DAIC controller are specified by the subscript α. Here, F_d represents the desired force.

In Fig. 4a, the desired force is a step signal with an amplitude of 20 N, where the sign indicates the direction only. In Fig. 4b, the desired force is a unit ramp signal. Both controllers achieve zero steady-state error under the step input. For the unit ramp input, when $\alpha = 1.0$, the steady-state error of DAIC is zero, whereas that of the impedance controller (IC) is -0.25 N, which is consistent with the analytical results in

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.

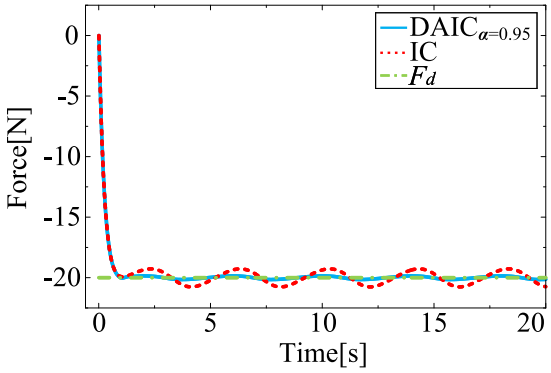
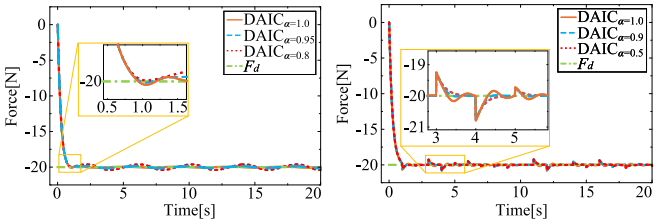


Fig. 5. Comparison of force tracking performance between DAIC and IC in dynamic environments.



(a) Continuous weak dynamic environment (b) Random strong dynamic environment

Fig. 6. Comparison of force tracking performance for DAIC with different parameters.

Appendix. The rise time is defined as the time required for the response to increase from 0% to 99% of the steady-state value. During the tracking process, the rise time of IC is 1.14 s with no overshoot, whereas DAIC achieves a rise time of 1.01 s with an overshoot of 0.274%. These results indicate that, in a static environment, DAIC exhibits a slightly faster response than the standard impedance controller, at the cost of a small overshoot.

Since static environments are idealized, environmental stiffness typically varies during real-world interactions. The force tracking performance of the two controllers is further evaluated in a dynamic environment, where the environmental stiffness is defined as $K_e = 400 + 40\sin(0.5\pi t)$ N/m, while all other parameters remain unchanged. The simulation results are presented in Fig. 5. The force tracking RMSE of the impedance controller (IC) is 0.269 N, whereas that of DAIC (with $\alpha = 0.95$) is 0.012 N. These results demonstrate that DAIC achieves significantly improved force tracking performance compared with the standard impedance controller under time-varying environmental stiffness, suggesting its effectiveness in practical applications.

To provide guidance for parameter selection in the practical implementation of DAIC, the influence of the parameter α on force tracking performance under dynamic environments is further investigated. Two operating conditions—continuous mild dynamic environments and randomly varying severe dynamic environments—are considered for evaluation.

For the case of smoothly and slowly varying environmental stiffness, the stiffness is defined as $K_e = 400 + 40\sin(0.5\pi t)$N/m. Three different values of α are considered,

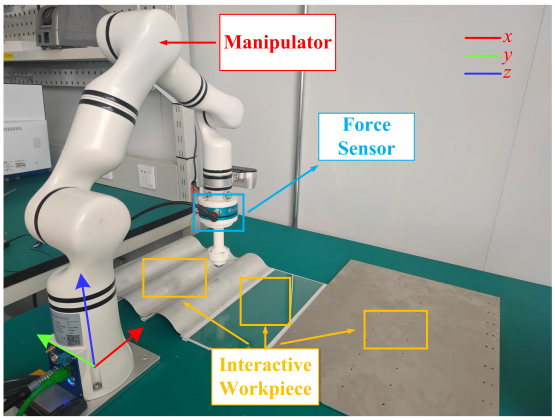


Fig. 7. Experimental setup.

and the simulation results are presented in Fig. 6a. The force tracking RMSE is 8.54×10^{-2} N for $\alpha = 0.8$, 1.19×10^{-2} N for $\alpha = 0.95$, and 1.58×10^{-3} N for $\alpha = 1.0$. These results indicate that increasing α enhances the controller's responsiveness to variations in environmental stiffness, leading to faster convergence of the force tracking error. However, since environmental stiffness in real-world interactions may not vary smoothly, a discontinuous and strongly random variation condition is further considered.

Under this condition, the environmental stiffness is defined as $K_e = 400 + 20p$ N/m, where p is a uniformly distributed random variable that is updated every second. A fixed random seed is used to ensure identical stiffness fluctuation patterns across all cases. The simulation results are presented in Fig. 6b. The force tracking RMSE is 2.37×10^{-2} N for $\alpha = 0.5$, 1.97×10^{-2} N for $\alpha = 0.9$, and 2.23×10^{-2} N for $\alpha = 1.0$. Compared with the previous operating condition, although the controller with $\alpha = 1.0$ exhibits a faster response, it suffers from pronounced overshoot, resulting in inferior overall performance compared with the case of $\alpha = 0.9$. Combining the results from both operating conditions, the selection of the parameter α involves a trade-off between force tracking smoothness and response speed. In practice, choosing $\alpha \approx 0.9$ provides a balanced performance. It should be noted that, in the present study, α is treated as a fixed design parameter rather than an online adaptive variable. This choice is made to preserve the simplicity and stability of the compensation recursion, since α directly determines the attenuation of accumulated force errors and affects the closed-loop memory effect. The adaptivity of the proposed DAIC mainly comes from the bounded time-varying gain $\beta(t)$, which is updated according to the force tracking error and its derivative. Nevertheless, developing an online adjustment law for α is a meaningful extension. For example, α may be reduced when large force overshoot or rapid error variation is detected, and increased when the force error becomes small and smooth, so as to further balance response speed and tracking smoothness. This topic will be investigated in future work.

B. Compliant Interaction Experiment on Plane

This subsection presents the implementation of the DAHIC algorithm in a real-world scenario, where comparative

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.

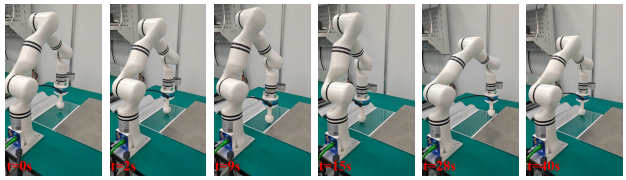
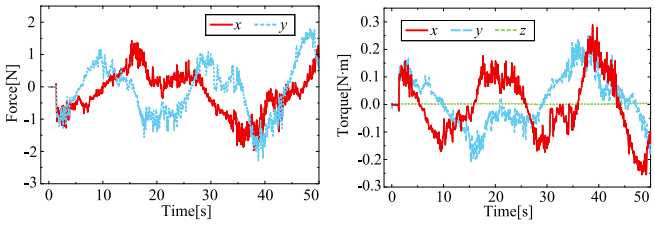
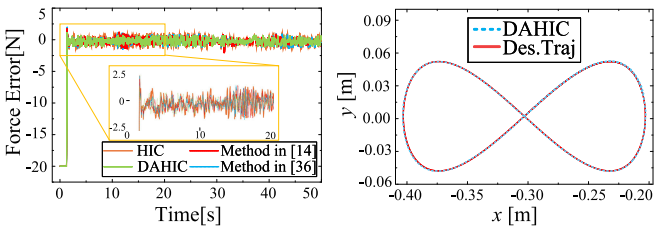


Fig. 8. Screenshot of the experimental procedure on plane.

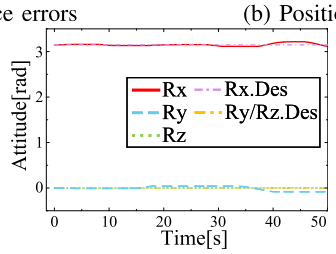


(a) Measured forces (b) Measured torques

Fig. 9. Measured forces and torques on plane.



(a) Force errors (b) Position errors



(c) Attitude errors

Fig. 10. Tracking errors on plane.

experiments are conducted to further validate its effectiveness and analyze the impact of controller parameters on tracking performance. The experimental setup is illustrated in Fig. 7. The proposed DAHIC algorithm is experimentally validated on a RealMan RM65-B-V robotic manipulator. A WRIST ST-6-170-6 force/torque (F/T) sensor is mounted between the EE and the manipulator to measure six-dimensional F/T data in the sensor coordinate frame. The transformation between the sensor frame and the task frame $\{T\}$ is constant. A control host receives F/T measurements and joint angle feedback via a local area network, computes control commands, and transmits them to the manipulator for execution.

In the experimental section, the six-DoF full-pose DAHIC algorithm is adopted. Constant-force contact is realized in the z-direction, while position-based impedance control is implemented in the x- and y-directions, as well as in the orientation DoFs. The impedance parameters for the translational DoFs (including both the force-controlled direction and the corresponding position-controlled directions) are set as $\boldsymbol{M}_d$ = diag(60, 60, 15) Ns^2/m, $\boldsymbol{B}_d$ = diag(300, 300, 1000) Ns/m, and $\boldsymbol{K}_d$ = diag(2000, 2000, 0) N/m. The impedance parameters for the orientation DoFs are $\boldsymbol{M}_o$ = diag(0.4, 0.4, 0.4) Nms^2/rad and $\boldsymbol{B}_o$ = diag(4, 4, 4) Nms/rad, while $\boldsymbol{K}_o$ is computed according to (11) with $\boldsymbol{K}'_o$ = diag(1, 1, 1) Nm/rad. The entire experimental procedure begins with a pre-contact phase, during which a certain distance is maintained between the EE (whose contact component is a bearing steel ball) and the acrylic plate surface. Owing to the persistent contact force error, $E(t)$ accumulates continuously until the EE eventually makes contact with the acrylic plate and converges to the desired force. The overall experimental procedure is illustrated in Fig. 8.

Figure 9 presents the forces and torques measured by the force sensor. It should be noted that the measured forces are confined to the x- and y-directions, while the force along the z-axis can be derived from the force tracking errors illustrated in Fig. 10. Therefore, it is not elaborated upon here. It is observable that, even for a planar contact surface with a constant normal direction, the coupled effects of normal pressure and contact-surface stiction render the forces and torques in the orthogonal directions non-negligible.

TABLE I

BENCHMARK COMPARISON OF FORCE TRACKING IN THE PLANAR CONTACT EXPERIMENT

Method	Force RMSE (N)	Max. Error (N)	Settling Time (s)
HIC	0.559	1.340	2.04
Method in [11]	0.434	1.479	1.89
Method in [17]	0.433	1.942	1.27
DAHIC	0.415	0.959	0.72

Figure 10 depicts the tracking errors across each DoF. Specifically, Fig. 10a compares the force tracking errors along the z-axis for the DAHIC and HIC algorithms. For the HIC algorithm, the RMSE is 0.561 N, with a maximum absolute error of 1.340 N; in contrast, the DAHIC algorithm achieves an RMSE of 0.414 N and a maximum absolute error of 0.959 N, corresponding to reductions of 26.2% and 28.4%, respectively. Notably, the maximum absolute error is computed starting from 0.5 s after initial contact, thereby excluding the overshoot occurring at the moment of impact.

To further benchmark the proposed method against recent adaptive interaction-control strategies, two additional controllers are implemented in the planar contact experiment: the variable admittance control method in [11] and the incremental hybrid impedance method in [17]. These two baselines are selected because they are closely related to adaptive compliance regulation and contact-force control under uncertain interactions. For a fair comparison, all controllers are evaluated using the same robot platform, desired trajectory, desired contact force, sampling period, and contact surface. The baseline controllers are adapted to the six-dimensional hybrid task-space structure used in this work, and their parameters are carefully tuned to obtain the best stable tracking performance. Due to space limitations, the force-error curves are added to Fig. 10a, while the quantitative force-tracking results are summarized in Table I. The settling time in Table I is defined

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.

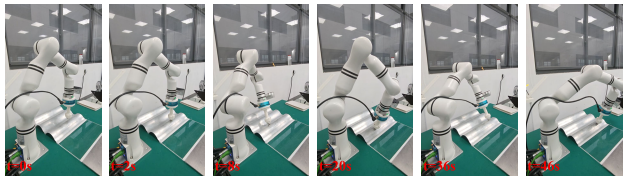
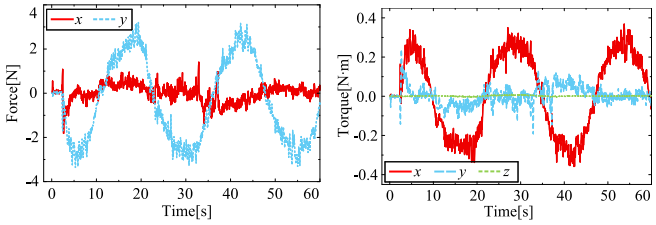


Fig. 11. Screenshot of the experimental procedure on curved surface.



(a) Measured forces (b) Measured torques

Fig. 12. Measured forces and torques on curved surface.

as the time after initial contact required for the force tracking error to enter and remain within 5% of the desired force magnitude (± 1 N in this experiment) for at least 1 s.

As shown in Table I, DAHIC achieves the lowest force-tracking RMSE, maximum absolute error, and settling time among the compared controllers. Compared with the selected recent adaptive baselines, the proposed direct force-compensation mechanism provides faster recovery after contact establishment and smaller sustained-contact force fluctuations. Therefore, the added benchmark further supports the improved force-tracking performance of DAHIC in the six-dimensional hybrid task-space experiment.

Since the primary objective of this experiment is normal-force regulation, the above benchmark focuses on force-tracking performance. The following analysis further presents the position and orientation tracking performance of the proposed DAHIC method to verify that the force-orthogonal motion and attitude regulation remain bounded during contact. Figure 10b illustrates the position tracking performance in the force-orthogonal directions during the planar compliant contact experiment, where "Des. Traj" denotes the desired trajectory. The RMSEs in the x- and y-directions are 0.577 mm and 0.437 mm, respectively. Figure 10c presents the tracking errors in the orientation DoFs, where "Rx", "Ry", and "Rz" represent the Euler angles about the x-, y-, and z-axes in the ZYX intrinsic rotation sequence. "Rx.Des" and "Ry/Rz.Des" denote the desired values for "Rx" and for "Ry/Rz", respectively. During the experiment, the RMSEs in the x-, y-, and z-directions are 0.0285 rad, 0.0482 rad, and 0.000645 rad, respectively.

C. Compliant Interaction Experiment on Curved Surface

To further validate the generalizability of the proposed algorithm across various contact surface geometries, the study extends the experimental scenario from planar surfaces to curved contact surfaces, thereby investigating the force tracking performance under conditions of rapidly varying surface curvature.

Since the z-axis of the EE coordinate frame is required to align with the normal direction of the contact surface during interaction with the workpiece, the EE frame is selected as the task coordinate frame $\{T\}$. Accordingly, the task space is decomposed into force-based impedance control along the z-axis of the EE frame and position-based impedance control along the orthogonal x- and y-axes. As the workpiece is a structured component with a known surface normal, position-based impedance control is also applied to the orientation DoFs. This control configuration enables compliant interaction by incorporating force feedback while tracking the desired motion commands, thereby reducing the risk of surface damage caused by uneven workpiece geometry or excessive controller stiffness. The overall experimental procedure is illustrated in Fig. 11.

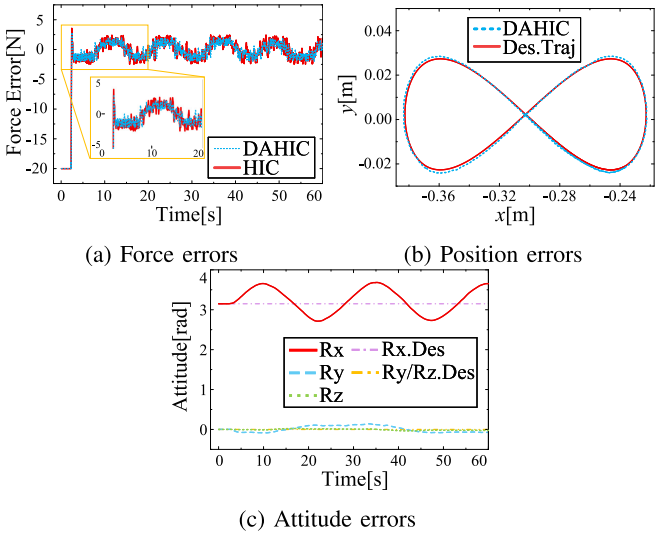
The impedance parameters for the translational DoFs are set as $\boldsymbol{M}_d = \text{diag}(60, 60, 10)$ Ns^2/m, $\boldsymbol{B}_d = \text{diag}(900, 900, 500)$ Ns/m, and $\boldsymbol{K}_d = \text{diag}(2000, 2000, 0)$ N/m. The impedance parameters for the orientation DoFs are $\boldsymbol{M}_o = \text{diag}(0.025, 0.025, 0.025)$ Nms^2/rad and $\boldsymbol{B}_o = \text{diag}(0.5, 0.5, 0.5)$ Nms/rad. $\boldsymbol{K}_o$ is designed as a zero matrix, which enables the EE orientation to be dynamically adjusted in response to the measured torque values. This mechanism allows the manipulator to adaptively conform to the surface curvature, thereby ensuring normal-direction contact throughout the interaction.

Since the EE coordinate frame is time-varying, the desired trajectory in the position subspace is defined in a fixed world coordinate frame. The desired velocity and position error feedback signals are transformed into the EE frame via a rotation matrix, and then the position-based impedance controller computes the control commands. Defining the desired trajectory in the world coordinate frame rather than in the EE frame offers the advantage of allowing intuitive judgment of physical reachability. It is worth noting that the desired trajectory in the world frame is three-dimensional. After transformation through the rotation matrix, only the motion components in the x and y directions are extracted for control command computation.

The measured forces from the force sensor are shown in Fig. 12, where the maximum forces in the y directions reach 3 N. This indicates that introducing impedance control in the directions orthogonal to the force subspace (x and y axes) is essential.

Figure 13 presents the tracking errors across each DoF. Specifically, Fig. 13a compares the force tracking errors along the z-axis for the DAHIC and HIC algorithms. For the HIC algorithm, the RMSE is 1.299 N, with a maximum absolute error of 2.483 N; in contrast, the DAHIC algorithm achieves an RMSE of 1.087 N and a maximum absolute error of 2.049 N, corresponding to reductions of 16.3% and 17.4%, respectively. Figure 13b illustrates the position tracking performance in the force-orthogonal directions during the curved-surface compliant contact experiment using the DAHIC algorithm. The RMSEs in the x- and y-directions are 0.337 and 0.780 mm, respectively. Owing to the increased complexity of force variations induced by surface curvature, the position tracking errors are observed to be larger than those in the planar contact scenario. Figure 13c depicts the tracking errors in the

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.



(a) Force errors (b) Position errors

(c) Attitude errors

Fig. 13. Tracking errors on curved surface.

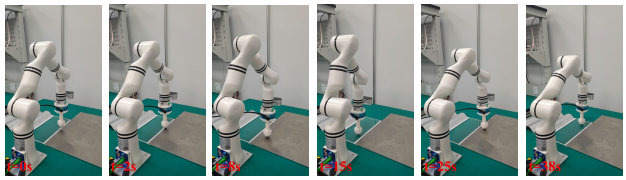


Fig. 14. Screenshot of the experimental procedure on different materials.

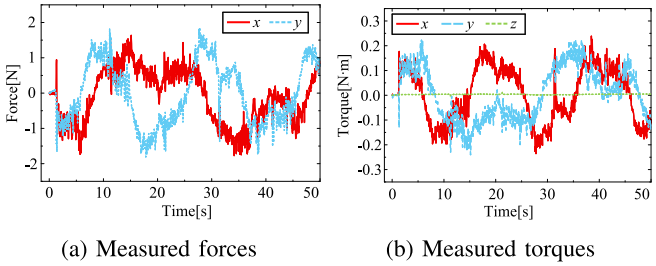
orientation DoFs. Since $\boldsymbol{K}_o$ is set to a zero matrix to ensure adaptability to the curved surface, the desired orientation values do not exert any actual influence on the experiment. As can be observed from the figure, owing to the periodicity of both the trajectory and the curvature variations, "Rx" exhibits periodic behavior and is capable of rapidly tracking the surface normal direction.

D. Compliant Interaction Experiment on Different Materials

To further validate the adaptability of the proposed algorithm to variations in contact surface material properties, experiments were designed involving frequent switching between two distinct contact surface materials, thereby enabling an assessment of the algorithm's comprehensive performance under abrupt changes in contact stiffness.

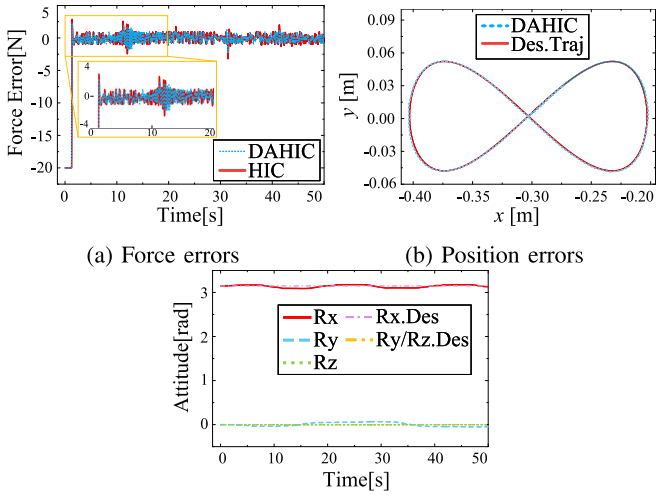
During this experiment, the manipulator's EE alternated reciprocally between an acrylic plate and an aluminum alloy plate. A snapshot of the experimental setup is shown in Fig. 14. The acrylic plate is composed of polymethyl methacrylate (PMMA), with an elastic modulus of approximately 2.4–3.4 GPa, whereas the aluminum alloy plate exhibits an elastic modulus of approximately 68.9 GPa. The substantial disparity in elastic moduli between the two materials poses a significant challenge to algorithm performance due to the abrupt transitions in contact stiffness.

The impedance parameters adopted in this experiment are consistent with those used in the planar contact experiment. As can be observed from Fig. 15, upon switching of the contact surface material, abrupt changes in contact force occur not only in the normal direction but also in the force-orthogonal



(a) Measured forces (b) Measured torques

Fig. 15. Measured forces and torques on different materials.



(a) Force errors (b) Position errors

(c) Attitude errors

Fig. 16. Tracking error on different materials.

directions, which markedly differs from the operating conditions of planar contact with a uniform material.

Figure 16 presents the tracking errors across each DoF. Specifically, Fig. 16a compares the force tracking errors along the z-axis for the DAHIC and HIC algorithms. Owing to the frequent switching of contact surface materials, the maximum absolute errors are substantially larger than those observed in the planar contact scenario with a single material. For the HIC algorithm, the RMSE is 0.693 N with a maximum absolute error of 3.211 N, while the DAHIC algorithm achieves an RMSE of 0.530 N and a maximum absolute error of 2.263 N, corresponding to reductions of 23.5% and 29.5%, respectively. It is evident that, although the frequent abrupt changes in contact stiffness somewhat degrade the force tracking performance, the DAHIC algorithm is capable of responding rapidly to sudden contact force variations and converging to a nominal level, in contrast to the HIC algorithm, which exhibits substantial oscillations around the desired contact force. Figure 16b illustrates the position tracking performance in the force-orthogonal directions during the contact-stiffness mutation experiment using the DAHIC algorithm, where the RMSEs in the x- and y-directions are 0.927 mm and 0.937 mm, respectively. Figure 16c depicts the tracking errors in the orientation DoF, with RMSEs in the x-, y-, and z-directions of 0.0317 rad, 0.0435 rad, and 0.00287 rad, respectively. The abrupt changes in contact stiffness do not induce any significant adverse effects on either the force-orthogonal direction

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.

or the orientation DoF, and the algorithm maintains stable and smooth convergence in these directions.

E. Phase-Wise Analysis of Contact Transition

To explicitly evaluate the controller behavior during different operational phases, the compliant interaction process is divided into three stages: free-space motion, contact establishment, and sustained contact. During free-space motion, the EE has not yet contacted the surface, and the measured contact force remains close to zero. Therefore, a large force tracking error exists, which drives the impedance-generated position correction until the EE approaches the contact surface. This behavior can be observed at the beginning of the force-error curves in Figs. 10a, 13a, and 16a.

The contact-establishment stage starts when the EE first touches the surface. In this short transition phase, the contact force rises rapidly, and a transient force overshoot may occur due to impact, finite servo bandwidth, and local contact stiffness. As shown in Figs. 10a, 13a, and 16a, both HIC and DAHIC exhibit an instantaneous force-error peak at the initial contact moment. However, after this impact interval, DAHIC suppresses the subsequent oscillation more effectively and recovers faster toward the desired contact force. This indicates that the proposed adaptive direct force-compensation mechanism improves the transient recovery behavior after contact establishment.

In the sustained-contact stage, the contact force fluctuates around the desired value, while the force-orthogonal position errors and attitude errors remain bounded. Therefore, the RMSE and maximum absolute error reported in the above experiments are mainly used to evaluate sustained-contact tracking performance after excluding the initial impact interval. In the planar experiment, for example, the maximum absolute error is computed starting from 0.5 s after initial contact. For the different-material experiment, the later abrupt force variations caused by material switching are retained in the evaluation because they reflect the controller's adaptability to contact-stiffness changes. Overall, the experimental results show that DAHIC improves not only sustained-contact accuracy, but also transient recovery after initial contact and after abrupt stiffness variations.

F. Effects of Position-Based Hybrid Impedance Control Parameters on Force Tracking

In the orthogonal direction (*i.e.*, position tracking), the tracking error of DAHIC ($\alpha = 0.9$) exceeds that of HIC. This may be attributed to the fact that the desired trajectory is designed based on an idealized structured contact surface with a fixed pose relative to the robot base. In practice, manufacturing tolerances often produce non-standard surfaces, and the surface pose may vary between experimental setups. As a result, even though force and position commands are decoupled at each time step, position commands can shift the contact location for the next step. On complex curved surfaces, small positional deviations can induce significant force fluctuations, affecting force feedback and subsequent control commands.

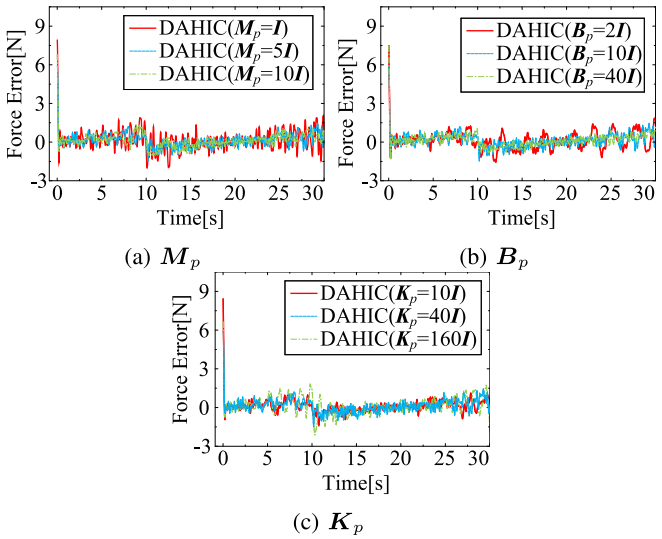


Fig. 17. Impact of position-based impedance control parameters on force tracking performance.

The proposed algorithm emphasizes accurate force tracking, while position control primarily facilitates motion orthogonal to the force subspace. Its strength lies in rapidly adapting to positional disturbances while maintaining desired force. Accordingly, position-based impedance stiffness should be properly limited, as overly stiff position control can conflict with force commands and cause undesirable oscillations.

To quantify this trade-off, a position RMSE threshold of approximately 1 mm in each force-orthogonal direction is adopted for the considered compliant contact tasks. The measured position RMSEs remain below this threshold in all experiments: 0.577/0.437 mm on the plane, 0.337/0.780 mm on the curved surface, and 0.927/0.937 mm on different materials in the x/y directions, respectively. Meanwhile, DAHIC reduces the force-tracking RMSE by 26.2%, 16.3%, and 23.5% in the three experiments. Therefore, the slight increase in position error is considered acceptable for the obtained improvement in force tracking.

Based on the above considerations in controller design, further experimental studies are conducted to investigate the influence of position-based impedance control parameters on force tracking performance. Let the impedance parameter submatrices corresponding to the position subspace be denoted as $\boldsymbol{M}_p$, $\boldsymbol{B}_p$, and $\boldsymbol{K}_p$, derived from $\boldsymbol{M}_d$, $\boldsymbol{B}_d$, and $\boldsymbol{K}_d$, respectively. Taking $\boldsymbol{M}_p = 5\boldsymbol{I}$ Ns^2/m, $\boldsymbol{B}_p = 10\boldsymbol{I}$ Ns/m, $\boldsymbol{K}_p = 40\boldsymbol{I}$ N/m as the baseline, the values of $\boldsymbol{M}_p$, $\boldsymbol{B}_p$, and $\boldsymbol{K}_p$ are varied individually to compare their effects on force tracking performance, as shown in Fig. 17. A quantitative comparison is presented in Table II.

The results indicate that increasing the inertia coefficient $\boldsymbol{M}_p$ and damping coefficient $\boldsymbol{B}_p$, while decreasing the stiffness coefficient $\boldsymbol{K}_p$, reduces the system's sensitivity to positional fluctuations. This, in turn, provides a wider admissible range for positional adjustment in the force-based impedance loop, thereby improving the overall compliance of the control system.

Actually, during experimental implementation, several practical difficulties were also encountered. First, the derivative of

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON OF FORCE TRACKING ERRORS UNDER DIFFERENT POSITION-BASED IMPEDANCE CONTROL PARAMETERS

Performance Metrics	Baseline	$\boldsymbol{M}_p = \boldsymbol{I}$	$\boldsymbol{M}_p = 10\boldsymbol{I}$	$\boldsymbol{B}_p = 2\boldsymbol{I}$	$\boldsymbol{B}_p = 40\boldsymbol{I}$	$\boldsymbol{K}_p = 10\boldsymbol{I}$	$\boldsymbol{K}_p = 160\boldsymbol{I}$
RMSE	0.543 N (100%)	0.763 N (140.5%)	0.537 N (98.8%)	0.684 N (125.9%)	0.520 N (95.7%)	0.508 N (93.5%)	0.672 N (123.7%)
MAE	1.458 N (100%)	2.047 N (140.3%)	1.379 N (94.5%)	1.845 N (126.5%)	1.193 N (81.8%)	1.388 N (95.1%)	2.119 N (145.3%)

the force tracking error was sensitive to measurement noise, which could cause excessive variation of the adaptive gain. To address this issue, filtered differentiation was used in the implementation. Second, the effective contact stiffness varied with the contact condition and surface state, which led to different transient force responses under the same nominal controller parameters. This issue motivated the use of the proposed direct force-compensation mechanism instead of explicit online environment identification. Third, the finite bandwidth of the inner-loop position servo introduced tracking lag during rapid transients. This effect was mitigated through conservative gain selection and by maintaining a clear separation between the force-controlled and position-controlled subspaces in the task frame.

V. CONCLUSION

This paper addresses the limited dynamic performance of traditional impedance control under time-varying environmental stiffness and proposes a direct force compensation–based adaptive hybrid impedance controller (DAHIC). The closed-loop stability and steady-state force tracking properties of the proposed controller are analytically investigated, and an adaptive parameter update law is derived. Simulation results demonstrate that the proposed direct force compensation strategy significantly improves the dynamic response of the impedance control system, achieving faster convergence and reduced force tracking errors.

To validate the proposed method in a multi-dimensional interaction scenario, comparative experiments are conducted in a six-dimensional task space by integrating position-based impedance control in the remaining DoFs. Experimental results show that DAHIC outperforms conventional hybrid impedance control, reducing the force tracking root mean square error by 12.5% and the maximum absolute error by 9.4%. In addition, appropriate tuning of the position-based impedance parameters is shown to effectively mitigate the influence of position subspace motion on force tracking performance, thereby improving overall system compliance.

Nevertheless, experimental observations indicate that delayed coupling effects between force-controlled and position-controlled subspaces may still introduce force fluctuations during interaction with complex curved surfaces. Future work will focus on improving the decoupling capability between force and position subspaces to further enhance both force tracking and position tracking performance in complex interaction tasks.

APPENDIX: SCALAR STEADY-STATE TRACKING INTERPRETATION OF DAIC

For additional insight into the tracking characteristics of the proposed direct force compensation adaptive impedance control (DAIC) mechanism, this appendix presents a scalar steady-state analysis under simplified nominal assumptions, including single-direction contact interaction, local linear stiffness, and idealized low-frequency servo response.

Under this scalar nominal setting, let $E(s)$ denote the Laplace transform of the position correction generated by the force-control channel, and let $F_d(s)$ and $\Delta F(s)$ denote the desired force and force tracking error, respectively. The corresponding scalar closed-loop tracking-error transfer function is defined as:

$$\psi(s) = \frac{\Delta F(s)}{F_d(s)} = -\frac{1}{1 + G(s)H(s)} \tag{41}$$

where $G(s) = E(s)/\Delta F(s)$ is the scalar transfer function of the force-control channel, and $H(s)$ represents the scalar environment model relating the position correction to the contact force. Under the simplified local linear contact assumption adopted here, $H(s) = K_e$. For comparison, DAIC denotes the proposed adaptive impedance controller and IC denotes the conventional impedance controller.

Based on the scalar transfer functions derived in the simplified analysis and (7), $\psi(s)$ can be written as:

$$\psi(s) = \begin{cases} \text{DAIC} : -\dfrac{M_d\alpha T s^3 + [M_d(1-\alpha) + B_d\alpha T]s^2 + B_d(1-\alpha)s}{\gamma_2(s)} \\ \text{IC} : -\dfrac{M_d s^2 + B_d s}{M_d s^2 + B_d s + K_e} \end{cases} \tag{42}$$

where

$$\begin{aligned} \gamma_2(s) = {} & M_d\alpha T s^3 + [M_d(1-\alpha) + B_d\alpha T]s^2 \\ & + [B_d(1-\alpha) + K_e(\alpha-\beta)T]s + (1-\alpha+\beta)K_e. \end{aligned} \tag{43}$$

For a unit step input, *i.e.*, $F_d(s) = 1/s$, the steady-state force tracking error is:

$$\Delta F_{ss} = \lim_{s\to 0} s\psi(s)F_d(s) = \lim_{s\to 0}\psi(s) = \begin{cases} \text{DAIC} : 0 \\ \text{IC} : 0 \end{cases}. \tag{44}$$

For a unit ramp input, *i.e.*, $F_d(s) = 1/s^2$, the steady-state force tracking error is:

$$\Delta F_{ss} = \lim_{s\to 0} s\psi(s)F_d(s) = \lim_{s\to 0}\frac{\psi(s)}{s} = \begin{cases} \text{DAIC} : -\dfrac{B_d(1-\alpha)}{K_e(1-\alpha+\beta)} \\ \text{IC} : -\dfrac{B_d}{K_e} \end{cases}. \tag{45}$$

Since

$$\frac{1-\alpha}{1-\alpha+\beta} < 1 \tag{46}$$

for $\alpha \in (0, 1]$ and $\beta > 0$, DAIC exhibits a smaller steady-state tracking error than the conventional impedance controller under a ramp input. In particular, when $\alpha = 1$,

$$\Delta F_{ss}^{\text{DAIC}} = 0. \tag{47}$$

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.

Therefore, in this simplified scalar interpretation, the proposed direct force compensation mechanism preserves zero steady-state error for constant force commands and improves the steady-state tracking performance for slowly varying force References. This appendix serves only as a supplementary scalar interpretation of the compensation mechanism, whereas the main theoretical result of the paper is the MIMO bounded-stability analysis given in Section III-D.

REFERENCES

[1] E.G. Ribeiro, R. Q. Mendes, M. H. Terra, and V. Grassi, "Second-order position-based visual servoing of a robot manipulator," *IEEE Robot. Autom. Lett.*, vol. 9, no. 1, pp. 207–214, Jan. 2024.

[2] H. Xing, Y. Xu, L. Ding, J. Chen, H. Gao, and M. Tavakoli, "Trajectory tracking control of wheeled mobile manipulators with joint flexibility via virtual decomposition approach," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 11808–11825, 2025.

[3] D. E. Whitney and J. M. Rourke, "Mechanical behavior and design equations for elastomer shear pad remote center compliances," *J. Dyn. Syst., Meas., Control*, vol. 108, no. 3, pp. 223–232, Sep. 1986.

[4] Y. Wei and Q. Xu, "Design of a new passive end-effector based on constant-force mechanism for robotic polishing," *Robot. Computer-Integrated Manuf.*, vol. 74, Apr. 2022, Art. no. 102278.

[5] C. Song, J. Wang, Q. Bu, J. Sun, and Q. Liu, "Design of a soft gripper with a changeable adapter for enhanced grasping and manipulation," *Robot Learn.*, pp. 1–21, Jul. 2026.

[6] C. H. Liu, C.-C.-A. Chen, and J.-S. Huang, "The polishing of molds and dies using a compliance tool holder mechanism," *J. Mater. Process. Technol.*, vol. 166, no. 2, pp. 230–236, Aug. 2005.

[7] Z. Hua, X. Rong, Y. Li, H. Chai, B. Li, and S. Zhang, "Analysis and verification on energy consumption of the quadruped robot with passive compliant hydraulic servo actuator," *Appl. Sci.*, vol. 10, no. 1, p. 340, Jan. 2020.

[8] Y. Wang et al., "Terminal sliding mode control of flexible end grinding force of wind turbine blade grinding robot," *Comput. Integr. Manuf. Syst.*, vol. 39, no. 5, pp. 35–243, 2020.

[9] N. Hogan, "Impedance control of industrial robots," *Robot. Computer-Integrated Manuf.*, vol. 1, no. 1, pp. 97–113, Jan. 1984.

[10] L. Ding, H. Xing, H. Gao, A. Torabi, W. Li, and M. Tavakoli, "VDC-based admittance control of multi-DOF manipulators considering joint flexibility via hierarchical control framework," *Control Eng. Pract.*, vol. 124, Jul. 2022, Art. no. 105186.

[11] H. Xing, Y. Liu, J. Chen, W. Li, L. Ding, and M. Tavakoli, "Variable admittance control for door opening with a wheeled mobile manipulator considering ground obstacles," *Intell. Service Robot.*, vol. 19, no. 2, pp. 1–15, Mar. 2026.

[12] M. H. Raibert and J. J. Craig, "Hybrid position/force control of manipulators," *J. Dyn. Syst., Meas., Control*, vol. 103, no. 2, pp. 126–133, Jun. 1981.

[13] M. T. Mason, "Compliance and force control for computer controlled manipulators," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-11, no. 6, pp. 418–432, Jun. 1981.

[14] H. Zheng, X. Zhai, H. Wu, J. Pan, Z. Xu, and X. Zhou, "Interaction model estimation-based robotic force-position coordinated optimization for rigid–soft heterogeneous contact tasks," *Biomimetic Intell. Robot.*, vol. 5, no. 1, Mar. 2025, Art. no. 100194.

[15] R. J. Anderson and M. W. Spong, "Hybrid impedance control of robotic manipulators," *IEEE J. Robot. Autom.*, vol. JRA-4, no. 5, pp. 549–556, Oct. 1988.

[16] Z. Li, W. Shang, and B. Zhang, "Hybrid impedance control of cable-driven parallel robots using adaptive friction compensation," *IEEE Trans. Ind. Electron.*, vol. 72, no. 1, pp. 703–713, Jan. 2025.

[17] Y. Zhang et al., "An incremental hybrid impedance algorithm for stable force—Position control of elastic materials," *IEEE Trans. Autom. Sci. Eng.*, vol. 23, pp. 3134–3144, 2026.

[18] X. Zhai et al., "Learning target-directed skill and variable impedance control from interactive demonstrations for robot-assisted soft tissue puncture tasks," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 5238–5250, 2025.

[19] H. Seraji and R. Colbaugh, "Force tracking in impedance control," *Int. J. Robot. Res.*, vol. 16, no. 1, pp. 97–117, 1997.

[20] T. Zhang, L. Jiang, S. Fan, X. Wu, and W. Feng, "Development and experimental evaluation of multi-fingered robot hand with adaptive impedance control for unknown environment grasping," *Robotica*, vol. 34, no. 5, pp. 1168–1185, May 2016.

[21] J. Duan, Y. Gan, M. Chen, and X. Dai, "Adaptive variable impedance control for dynamic contact force tracking in uncertain environment," *Robot. Auto. Syst.*, vol. 102, pp. 54–65, Apr. 2018.

[22] X. Liang, H. Zhao, X. Li, and H. Ding, "Force tracking impedance control with unknown environment via an iterative learning algorithm," in *Proc. 3rd Int. Conf. Adv. Robot. Mechatronics (ICARM)*, Jul. 2018, pp. 158–164.

[23] H. H. Li, K. R. Hao, and Y. S. Ding, "Force/Position cooperative control for 6-DOF parallel platform," *Adv. Mater. Res.*, vols. 479-481, pp. 1790–1794, Feb. 2012.

[24] L. Gracia, J. E. Solanes, P. Muñoz-Benavent, J. Valls Miro, C. Perez-Vidal, and J. Tornero, "Adaptive sliding mode control for robotic surface treatment using force feedback," *Mechatronics*, vol. 52, pp. 102–118, Jun. 2018.

[25] L. Rojas-García, I. Bonilla-Gutiérrez, M. Mendoza-Gutiérrez, and C. Chávez-Olivares, "Adaptive force/position control of robot manipulators with bounded inputs," *J. Mech. Sci. Technol.*, vol. 36, no. 3, pp. 1497–1509, Mar. 2022.

[26] P. Chen, H. Zhao, X. Yan, and H. Ding, "Force control polishing device based on fuzzy adaptive impedance control," in *Proc. Int. Conf. Intell. Robot. Appl.*, 2019, pp. 181–194.

[27] H. Xing, L. Ding, J. Chen, H. Gao, and Z. Deng, "Robotic valve turning with a wheeled mobile manipulator via hybrid passive/active compliance," *Sensors*, vol. 24, no. 17, p. 5559, Aug. 2024.

[28] F. J. Abu-Dakka and M. Saveriano, "Variable impedance control and learning—A review," *Frontiers Robot. AI*, vol. 7, Dec. 2020, Art. no. 590681.

[29] Y. Sun, Z. Peng, J. Hu, and B. K. Ghosh, "Event-triggered critic learning impedance control of lower limb exoskeleton robots in interactive environments," *Neurocomputing*, vol. 564, Jan. 2024, Art. no. 126963.

[30] X. Zhang, L. Sun, Z. Kuang, and M. Tomizuka, "Learning variable impedance control via inverse reinforcement learning for force-related tasks," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 2225–2232, Apr. 2021.

[31] J. Luo, E. Solowjow, C. Wen, J. A. Ojea, and A. M. Agogino, "Deep reinforcement learning for robotic assembly of mixed deformable and rigid objects," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2018, pp. 2062–2069.

[32] Y. Zhang, L. Cheng, R. Cao, H. Li, and C. Yang, "A neural network based framework for variable impedance skills learning from demonstrations," *Robot. Auto. Syst.*, vol. 160, Feb. 2023, Art. no. 104312.

[33] J. Buchli, F. Stulp, E. Theodorou, and S. Schaal, "Learning variable impedance control," *Int. J. Robot. Res.*, vol. 30, no. 7, pp. 820–833, 2011.

[34] Y. Ding, J. Zhao, and X. Min, "Impedance control and parameter optimization of surface polishing robot based on reinforcement learning," *Proc. Inst. Mech. Eng., B, J. Eng. Manuf.*, vol. 237, nos. 1–2, pp. 216–228, Jan. 2023.

[35] T. Sun and J. Yang, "Adaptive interaction control of compliant robots using impedance learning," *Sensors*, vol. 22, no. 24, p. 9740, Dec. 2022.

[36] P. Xu, Z. Li, X. Liu, T. Zhao, L. Zhang, and Y. Zhao, "Reinforcement learning-based distributed impedance control of robots for compliant operation in tight interaction tasks," *Eng. Appl. Artif. Intell.*, vol. 136, Oct. 2024, Art. no. 108913.

[37] A. Peer, "Design and control of admittance-type telemanipulation systems," *Ph.D. dissertation, Dept. Elect. Eng. Inf. Technol., Technische Universität München*, Munich, Germany, 2008.

[38] H. Xing et al., "Multi-objective optimization with priority-based execution for mobile manipulators: An HQP approach integrating adaptive motion planning and obstacle avoidance," *Robot. Auto. Syst.*, vol. 201, Jul. 2026, Art. no. 105457.

[39] H. Xing et al., "Inverse kinematics and redundancy resolution in manipulators: Methods, optimization objectives, and future trends," *Biomimetic Intell. Robot.*, vol. 6, no. 2, Jun. 2026, Art. no. 100339.

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on November 06,2026 at 09:21:08 UTC from IEEE Xplore. Restrictions apply.